India SMART UTILITY Week 2026

GIFT CITY District Cooling System Infrastructure

GUJARAT INTERNATIONAL FINANCE TEC-CITY

"GROWTH PROJECTION OF INDIA'S COOLING DEMAND"

- ❑ **Substantial growth in next 15 Years in Space cooling demand.**
- ❑ **Cooling is necessity and no longer a luxury service.**

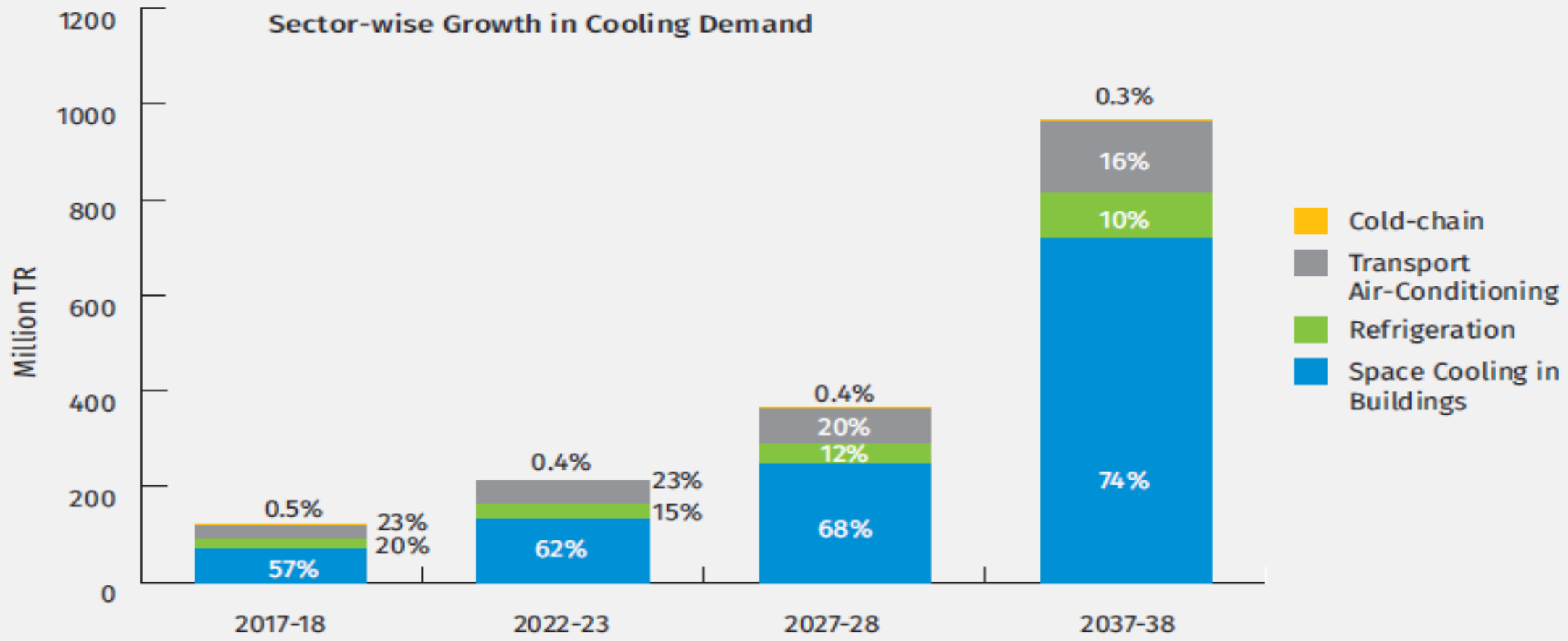
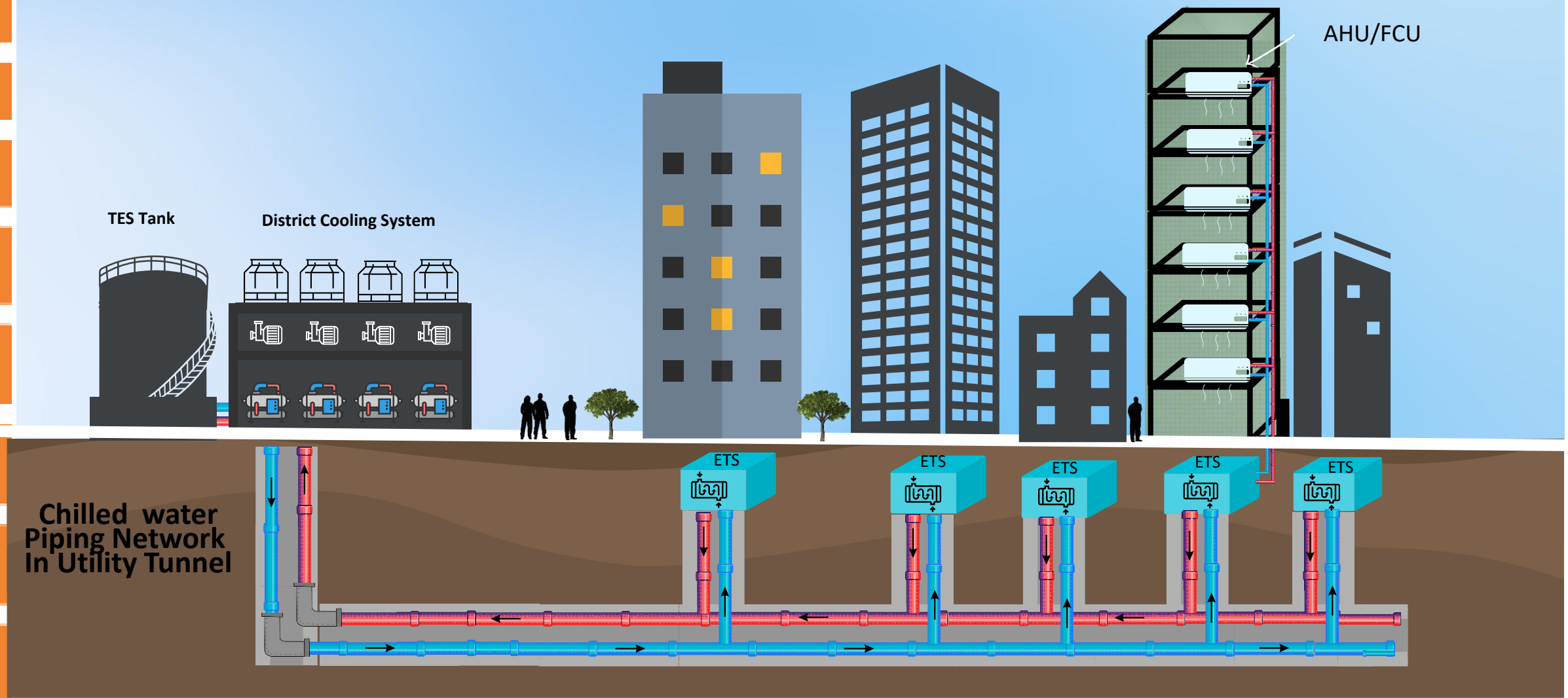


Figure B: Sector-wise Growth in Cooling Demand

Source- INDIA COOLING ACTION PLAN (ICAP)

DITRICT COOLING SYSTEM OVERVIEW

- A central cooling plant generates the cooling-energy demand of various mix use buildings
- Distribution of cooling energy via. Preinsulated piping network in utility tunnel
- Cooling energy transfer (ETS) stations with BTU metering and control valves



WHY DISTRICT COOLING SYSTEM

Energy savings

- 20-30% reduction in energy consumption
- 50% Peak power demand reductions
- Load balancing through TES Tanks
- Series counterflow high delta T operations

Economy of scale

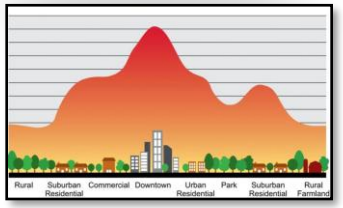
- Less installed capacity due to diversity
- Centralized operations
- Integration of multiple plants
- Ring main piping network

End-users

- Less tariff
- Comfort
- Reliability
- Easy to maintain

Sustainability

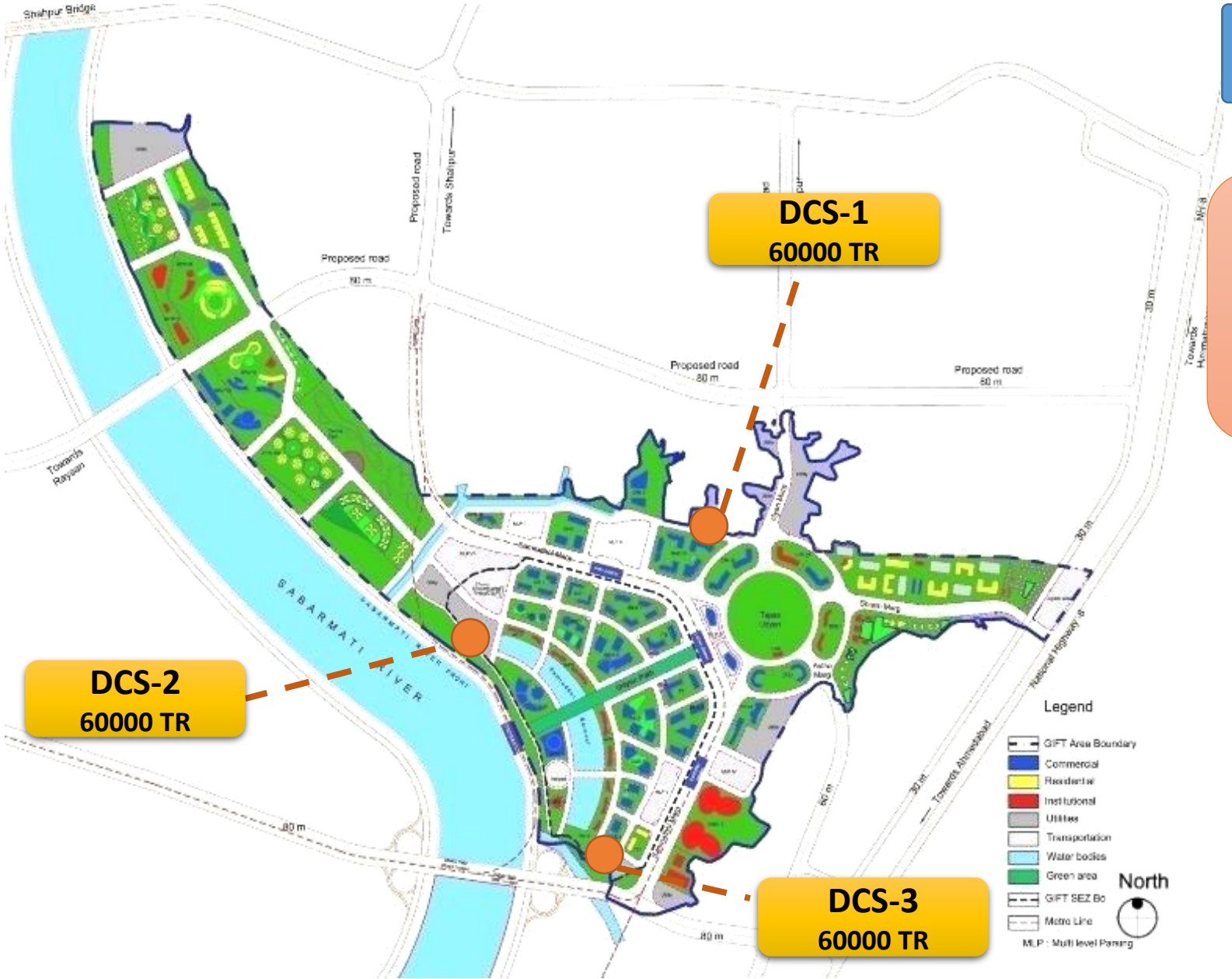
Reduced CO2 emissions



Eliminate heat islands

Reduced Global warming

GIFT MASTER PLAN-DCS



TOTAL BUA : 62 mn.sq.ft
Total DCS LOAD : 1,80,000 TR

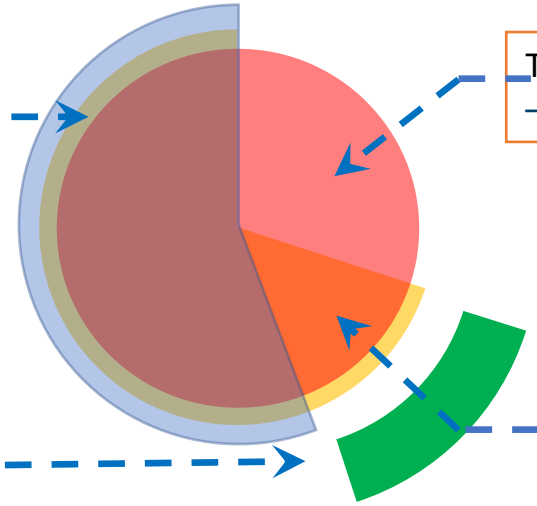
AREA BIFURCATION:

COMMERCIAL : 67% (42 mn.sq.ft.)
RESIDENTIAL : 22% (14 mn.sq.ft.)
SOCIAL : 11% (6 mn.sq.ft.)

MIXED USE DEVELOPMENTS:

1. GRADE A OFFICE SPACES
2. RESIDENTIAL APARTMENTS
3. HOTELS
4. HOSPITAL
5. DATA CENTRE
6. UNIVERSITIES
7. UTILITY BUILDINGS
8. RETAIL/MALL/SOCIAL

DCS-CITY LEVEL CAPACITY PLAN

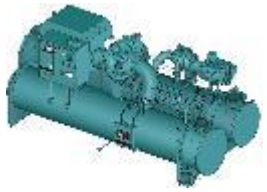


Total Chiller Installation – **1,50,000 TR**

Total Cooling capacity – **2,40,000 TR**

Thermal Energy Storage Capacity **-30,000 TR**

Diversified HVAC Capacity = **1,80,000 TR**



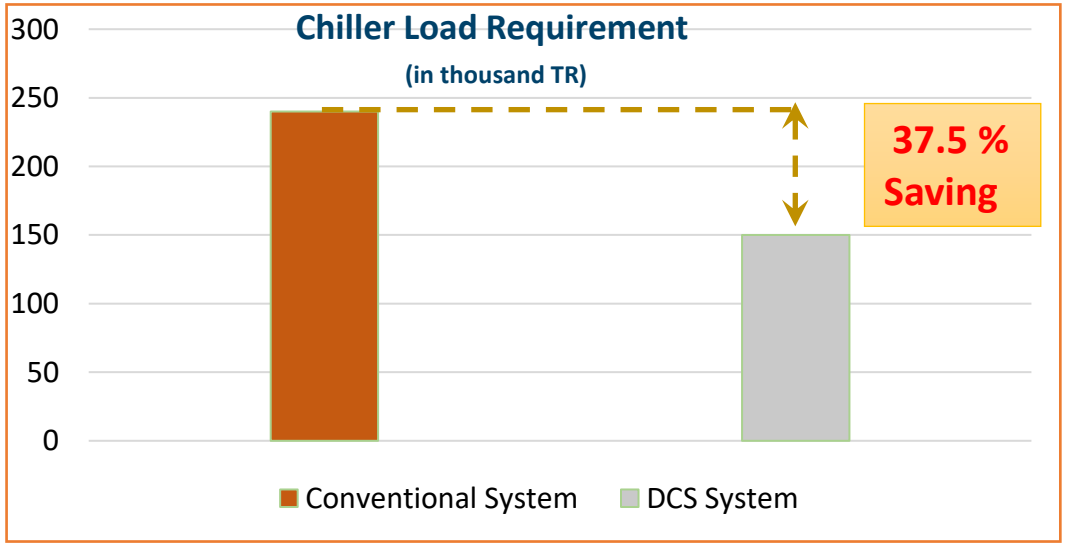
1,50,000 TR

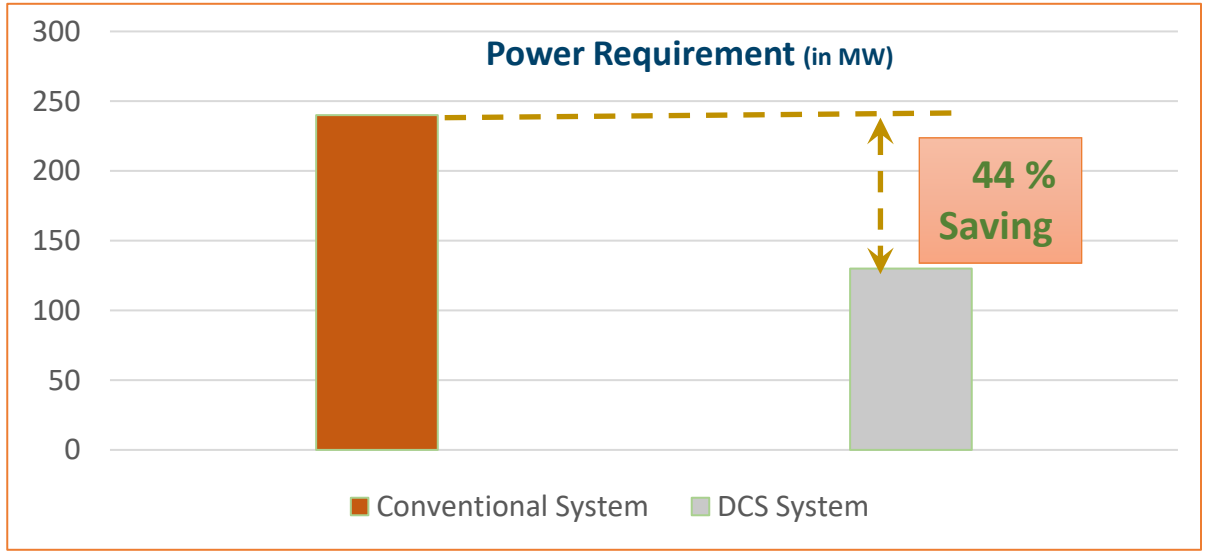
+

30,000 TR

=

DCS Generation Capacity= 1,80,000 TR





GIFT DCP-1 UNIQUE FEATURES

- Series counterflow chillers with high 9 Deg C Delta T
- Stratified TES tank for load levelling & Peak shaving
- 24X7 Operations monitored with Central Command center
- Pre insulated distribution pipe network in tunnel
- Precise flow control through PIBCV control valves
- 3D Tracer system for automatic dosing of chemicals
- Reuse of STP treated water for cooling tower make up

DCP-1 PLANT :

10,000 TR Operational since 2015

10,000 TR under Construction

CUSTOMERS:

22+ Buildings connected

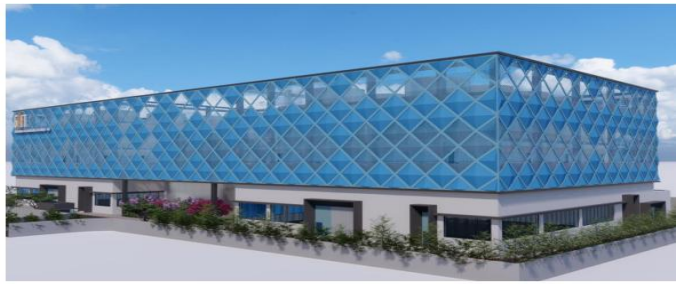
50+ Buildings under progress

PIPING NETWORK:

12+ Kilometers of Piping network operational

15+ Kilometers under progress

Isometric view from North-West Direction



Isometric view from South-West Direction

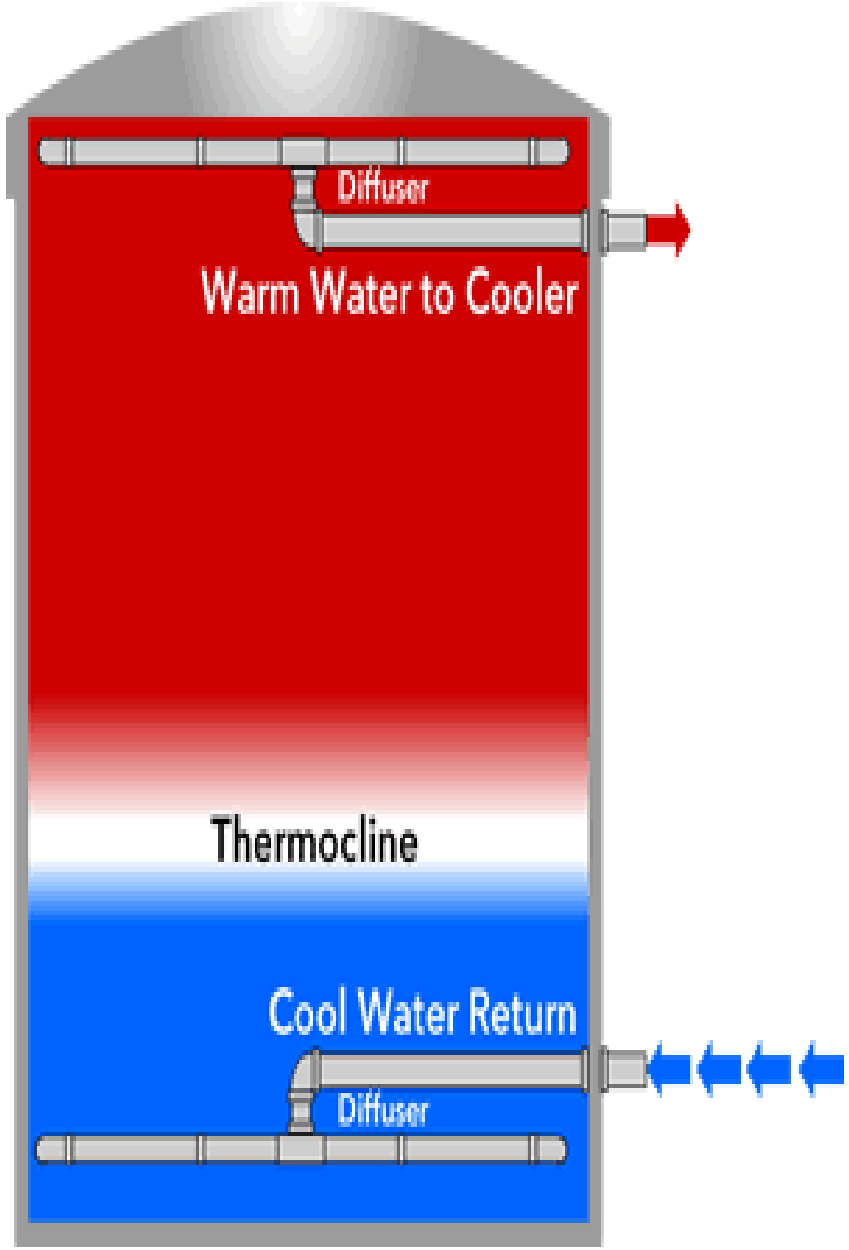
DCP-2 PLANT :

10,000 TR under tendering process

CHW PIPING NETWORK IN UYILITY TUNEL-OVERVIEW

Description	Total planned	Operational
DTA Area	30 KM	8 KM
SEZ AREA	25 KM	4 KM

TES Tank – An important asset of DCS



Key Highlights :

- **Insulated tank with provision of thermal sensors**
- **Stores thermal energy in the form of chilled water**
- **charged by running the chiller at best efficiency point at all times**
- **Load leveling to cut off peaks**
- **Run TES during night times while chiller is off**
- **Can be used as back up for 4-5 hours in case of chiller power failure**
- **Can be used as fire fighting storage in case of emergency**

DCS load projection and analysis is carried out considering total 28.24 Mn . Sq. feet of committed Built up area till date.

Details of BUA and connected DCS load is as follows;

Sr No.	Building Status	DTA BUA (Mn. Sq.feet)	SEZ BUA (Mn. Sq.feet)	Total BUA (Mn. Sq.feet)
1	Operational	3.65	2.01	5.66
2	Under Construction	10.67	8.83	19.50
3	Under Planning	NA	3.08	3.08
	Total BUA	**14.32**	**13.92**	**28.24**

Sr No.	Building Status	DTA Load (TR)	SEZ Load (TR)	Total Load (TR)
1	Operational	10977	6028	17005
2	Under Construction	33210	24202	57412
3	Under Planning	NA	9730	9730
	Total DCS LOAD (TR)	**44187**	**39960**	**84147**

"DCS COMPLETE ECO SYSTEM IN GIFT CITY"

ALLOTMENT	• Allot BUA as per masterplan
PLANTS	• Build DCP in phases
NETWORK	• Lay CHW piping network in Tunnel
ETS STATIONS	• Build ETS stations at building level

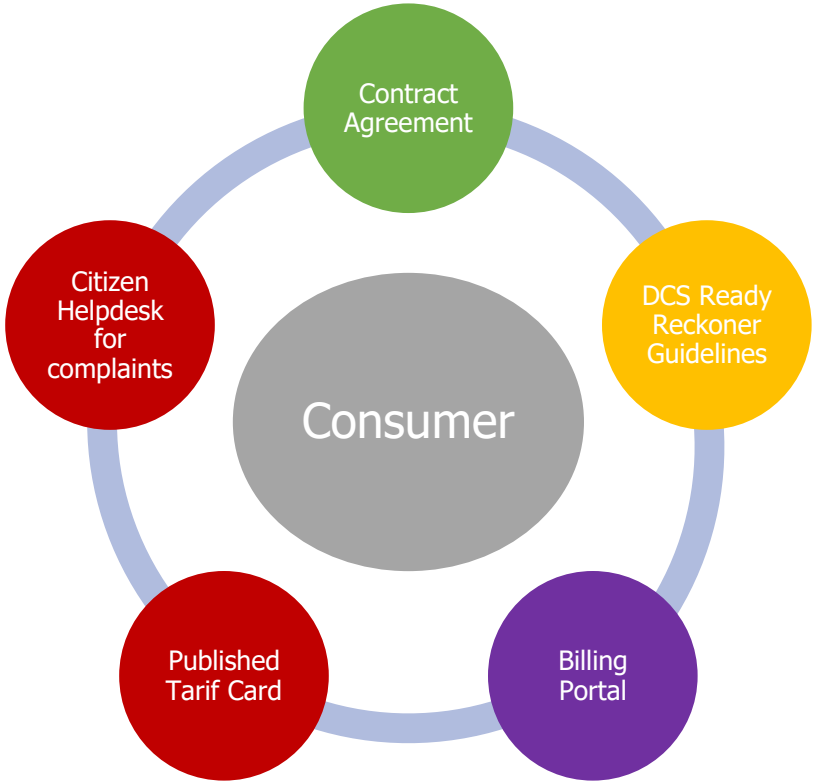
DCS INFRA. DEVELOPMENT

Development Permission
- Space Plan
- Utility interface

Construction Clearance
- Design Basis
- GFC drawings

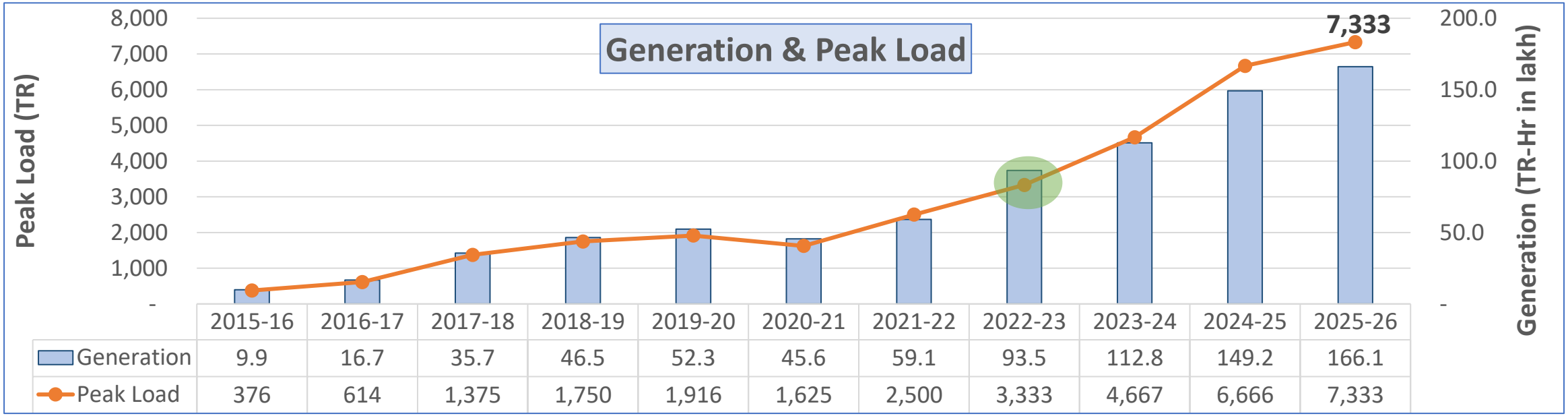
Occupancy Certificate
- Periodic interim audits
- Commissioning validation

BUILDING DEVELOPER INTERFCAE

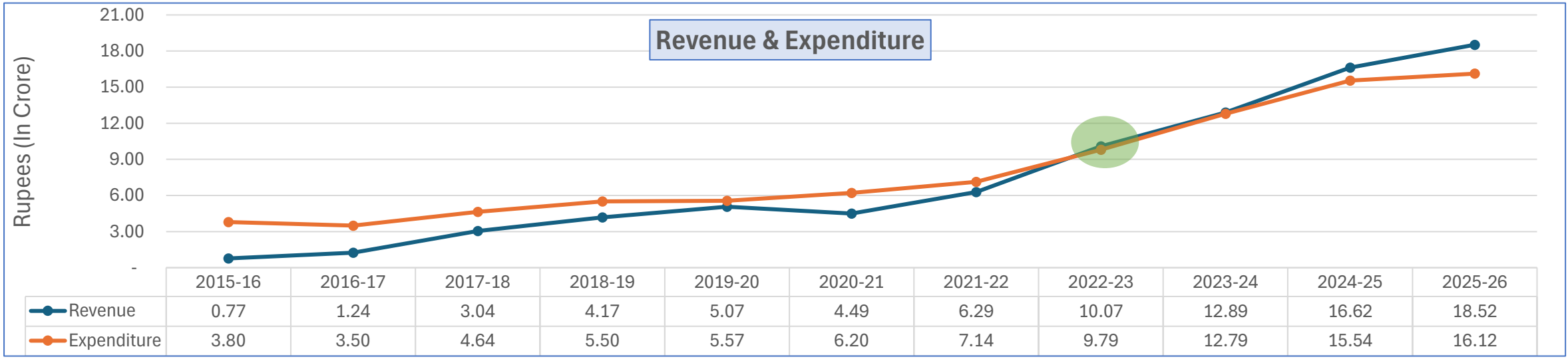


CONSUMER CARE

Operational Trends



	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26
Generation	9.9	16.7	35.7	46.5	52.3	45.6	59.1	93.5	112.8	149.2	166.1
Peak Load	376	614	1,375	1,750	1,916	1,625	2,500	3,333	4,667	6,666	7,333



	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26
Revenue	0.77	1.24	3.04	4.17	5.07	4.49	6.29	10.07	12.89	16.62	18.52
Expenditure	3.80	3.50	4.64	5.50	5.57	6.20	7.14	9.79	12.79	15.54	16.12

"DCS IN INDIA-CHALLENGES AND ACTIONS POINTS"

Category	Challenge	Mitigation
Business Forecast	Very low loads during initial 3-4 years as against forecasted. Load variability Day vs night	Effective use of Thermal energy storage tank to minimize chiller part load operations
Technical	Issues in Buildings internal HVAC design	Published design guidelines for building developers and created review mechanism to align to DCS system
Operational	Incorrect Operation at building level	Created awareness, training of building facility operators conducted periodically
Operational	Low delta T syndrome	Use of PIBCV control valves for effective control on DCS side and BMS controlled operation in building side Mandate high delta T indoor units
Operational	Water quality issues	Continuous monitoring, implemented automatic 3D tracer system
Technical	Energy losses in distribution network	Implemented stare of art factory spray insulation of piping and TES Tank. Supply Temperature deviation @ ±0.5 Deg C is maintained
Financial / Business	Revenue leakage (production vs billed mismatch)	During initial years due to low load conditions, the BTU meter under recording is observed. WIP to minimize this by sizing BTU meter for higher velocities or installing small meter for initial phase.
Financial / Business	Customer acceptance -DCS-Cooling as a Service-System Awareness	Mandate for DCS in GIFT City Competitive tariffs compared to conventional systems
Resource	Availability of key equipment and its spares in the country	Key tech. equipment and products sourced from other countries, spares management dedicated of GIFT site
Policy Level	No regulations/policy exists in the country	Created inhouse policy document inline with best international practices

"DCS IN INDIA-CHALLENGES AND ACTIONS POINTS"

Indian Consumer Behavior towards DCS

- Increase Awareness through marketing, publications, education by all stakeholders
- Residential users must be enlightened about life cycle cost benefits of DCS. Case studies to be published

Regulatory Framework (Tariff & Consumer Care)

- Government to establish "Energy Authority" similar to electricity regulatory for District cooling
- Guidelines, SOP, codes to be revisited/amended for developers to ensure proper installations in their buildings

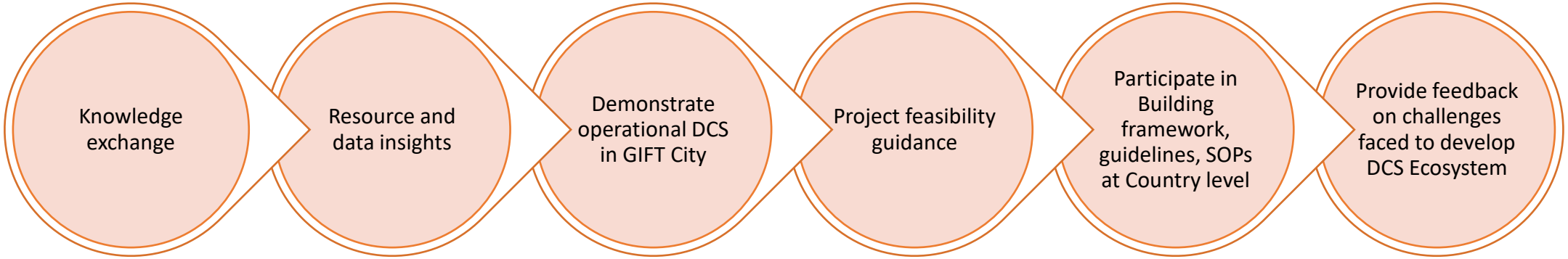
Investment/Funding for DCS

- Financial institutions to explore probable SOPs about funding DCS projects,
- Government to provide subsidies, incentives, tax benefits to promote DCS as a viable financial model

Retrofitting existing HVAC system in buildings to DCS compatible systems

- Systematic planning to switchover without shutdown.
- Awareness through energy auditing of existing system in the building compared with switching to DCS system

"HOW GIFT CAN CONTRIBUTE TO INDIA'S DCS JOURNEY"



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THANK YOU